INVERTIS UNIVERSITY BAREILLY
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BME-102/BME-202 – Basic Civil &Mechanical Engineering

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Course Objectives:
The objectives of this Course is to provide-

1. An introductory treatment of *Basic Mechanical Engineering* to all the students of engineering.
2. A working knowledge of statics with emphasis on force equilibrium.
3. To provide an understanding of the kinds of stress and deformation and how to determine them in a wide range of simple, practical structural problems
4. To introduce the students the various manufacturing processes carried out in various industries and properties on materials used.
5. To understand the basics of energy interactions.
6. A working knowledge of IC Engines, refrigerator, air conditioner etc.

MODULE -1

Introduction to Mechanics of Solid:
Normal and shear Stress, strain, Hookes' law, Poisson's ratio, elastic constants and their relationship, stress-strain diagram for ductile and brittle materials, factor of safety. Basic Numerical problems,
Force, force system, conditions of equilibrium, calculations of the reactions, moment, Types of beams under various loads, Statically Determinate Beams, Shear force and bending moment in beams, Shear force and bending moment diagrams, Basic Numerical problems,

MODULE- 2

Introduction to Thermal Engineering:
Thermodynamics: Thermodynamic systems, State & properties, Thermodynamic equilibrium & processes, Heat & work, Work done for different polytrophic processes, Zeroth law of thermodynamics and its applications, First law of thermodynamics, Steady flow energy equation, Application of first law to various thermodynamic systems and its limitations, concept of heat engine, refrigerator and heat pump, Carnot cycle, Second law of thermodynamics and its statements, Concept of entropy.
IC Engine: Basic Components, Construction and Working of Two stroke and four stroke SI & CI engine, merits and demerits, scavenging process; Introduction to hybrid vehicles.
Refrigeration and Air-Conditioning: Meaning, working of a simple refrigerator, meaning and application, humidity, dry bulb, wet bulb, and dew point temperatures, construction and working of window air conditioner.

MODULE -3

Introduction to Manufacturing Processes:
Engineering Materials and their properties: Metal, non-metal, ferrous, non-ferrous materials, strength, ductility, malleability etc.
Manufacturing Processes: Positive, negative and zero process, casting, joining processes, machining, forming.

After the successful completion of the course the students will be able (Course Outcomes)-

CO1	To define force, stress-strain, factor of safety, beams and various laws of thermodynamics.
CO2	Understand the basic laws of thermodynamics and their applications in
CO3	Apply laws of thermodynamics on different SFEE devices, concepts of refrigeration for calculation of COP, conditions of equilibrium compute load and reactions on different types of beams.
CO4	To differentiate various beams and thermodynamics systems.
CO5	To judge the feasibility of a thermodynamic process and Carnot cycle.
CO6	Develop basic knowledge and awareness of various manufacturing processes.

TEXT BOOKS

1. Engineering Mechanics: Ramamrutham S., 17th edition New Delhi : Dhanpat Rai Publications , 2002

2. Engineering Mechanics: S. S. Bhavikatti & Rajshekhrappa, 2nd edition: New Age International Publishers, 2012

3. Engineering Mechanics : Meriam J. L., 5th ed. Singapore : John Wiley & Sons (ASIA) Pte. Ltd., 2003

4. Engineering Mechanics: Statics & Dynamics: Shames Irving H.- 4th ed. New Delhi : Prentice Hall of India , 1998 to 2006

5. Engineering with Experiments : Dr. D. S. Kumar, S. K. Kataria & Sons., 2015, 12, 08

6. A Text Book of Engineering Mechanics: R. S. Khurmi, S. Chand Publication, 2004, 2011

7. A Text Book of Engineering Mechanics: R. K. Rajput, Dhanpat Rai Publication, 2004

8. Engineering Mechanics: With Hints To All Tutorial Problems: Kumar, K. L, 3rd .Rev. Ed. New Delhi : Tata McGraw Hill, 2005

9. Engineering Thermodynamics: P. K. Nag, TMH Publication.

10. A Textbook of Manufacturing Processes: Workshop Technology, R.S. Khurmi, S.Chand Publication, 2008

11. Basic Mechanical Engineering: Dr. D. S. Kumar, S K Kataria & Sons-New Delhi.

12. Thermodynamics "An Engineering Approach": Y. A. Cengel and Boles, McGraw Hills Publication, 8th Edition, 2016.